

Verifying a Simple DAG Scheduler

(TODO: My ID here) - COMP440 Assignment One

Aims and Objectives

The objective of this report is to model and prove the termination of a directed acyclic graph (DAG) scheduler in Idris2 and Coq (the precise theorem will be stated at the end of the next section). The proof will then be used as a vehicle to compare both languages by considering their available tooling (specifically interactive editing features, content of the standard library, and package managers), approaches to encoding propositions and proofs, as well as their methods for judging these propositions. Finally, a subjective account of both languages' developer experience will be given.

Preliminary Models, Definitions, and Proofs

Definition 1

A "Scheduler state" over a directed graph $G = (V, E)$ is comprised of four sets of tasks that partition V : 1. The set of finished tasks 2. The set of ready tasks 3. The set of pending tasks 4. The set of running tasks, which may contain at most one element.

For brevity, this construct may occasionally be referred to as a "scheduler."

Definition 2

A scheduler state over $G = (V, E)$ is "well formed" if for every task t in the "finished" set, every element d such that $d \in t$ holds is in the "finished" set.

Definition 3

A task t is enabled in a scheduler state S over a directed graph $G = (T, E)$ if t is in the “pending” set of S , and for all d in T such that dEt holds, d is in the “finished” set of S .

Definition 4

A “step” or “state transition” is a 2-tuple of scheduler states (S_1, S_2) over a directed graph $G = (T, E)$ such that exactly one of the following conditions hold: 1. The set of “running” tasks in S_1 is empty, and there exists a task t such that t is in the “ready” set of S_1 and the “running” set of S_2 . 2. There exists a task t such that t is in the “running” set of S_1 , t is in the “finished” set of S_2 , and the “running” set of S_2 is empty. 3. There exists a task t such that t is enabled in S_1 and t is in the “ready” set of S_2 .

Definition 5

A “trace” between two scheduler states S_1 and S_2 is a witness of $S_1 \text{ Step}^* S_2$, where Step^* is the transitive-reflexive closure of the step relation.

Definition 6

a scheduler state S is “reachable” from a starting scheduler state S_0 if there is a trace from S_0 to S . The set $\text{Reach}(S_0)$ are all states that are reachable from S_0 .

Definition 7

A scheduler state S is “terminal” if there does not exist an S' such that $S \text{ Step } S'$ holds.